

9.13 By the same consideration as in Prob. 9.12, the magnetization density is $\vec{m} = \hat{z} M / \frac{4}{3} \pi R_0^3$.

Since $\vec{r} \times \vec{m}$ is tangent to the sphere, $\nabla \cdot (\vec{r} \times \vec{m})$ will be zero on the surface and therefore electric multipole moments are all zero. For magnetic multipole moments, only M_{10} will be non-zero and $\nabla \cdot \vec{m}$ is the magnetization surface density and is $\vec{m} \cdot \vec{n} = m \cos \theta$ on the surface.

Then, $M'_{10} = - \int r^2 Y_{10}^* m \cos \theta \delta(r - R_0) d^3x$

Only $m=0$ will contribute to the integral and $Y_{10}^*(\theta, \phi) = \sqrt{\frac{3}{4\pi}} P_1(\cos \theta)$. Therefore,

$$\begin{aligned} M'_{10} &= - \sqrt{\frac{3}{4\pi}} m R_0^{1+2} \int [1 + \beta P_2(\cos \theta)]^{1+2} P_1(\cos \theta) \cos \theta d(\cos \theta) \\ &= - \sqrt{\frac{3}{4\pi}} 2\pi m R_0^{1+2} \int_{-1}^1 [1 + \beta P_2(\cos \theta)]^{1+2} P_1(\cos \theta) \cos \theta d(\cos \theta) \end{aligned}$$

To the lowest order in β , when $\lambda=0$, we have

$$M'_{00} = - \sqrt{\frac{1}{4\pi}} 2\pi m R_0^2 \int_{-1}^1 2\beta P_2(\cos \theta) \cos \theta d(\cos \theta) = 0$$

For $\lambda=1$, then

$$\begin{aligned} M'_{10} &= - \sqrt{\frac{3}{4\pi}} 2\pi m R_0^3 \int_{-1}^1 3\beta P_2(\cos \theta) P_1(\cos \theta) \cos \theta d(\cos \theta) = - \sqrt{\frac{3}{4\pi}} 6\pi m R_0^3 \beta \int_{-1}^1 \frac{1}{2}(3x^2-1)x^2 dx \\ &= - \sqrt{\frac{3}{4\pi}} 3\pi m R_0^3 \beta \cdot \frac{8}{15} = - \sqrt{\frac{3}{4\pi}} \frac{3\pi R_0^3 \beta M}{\frac{4}{3}\pi R_0^3} \frac{8}{15} = - \sqrt{\frac{3}{4\pi}} \frac{6}{5} M \beta \end{aligned}$$

Now, using Eq. (9.151), the power radiated per unit solid angle is

$$\begin{aligned} \frac{dP(l, \nu)}{d\Omega} &= \frac{Z_0}{2k^2} |a_{nl}(l, \nu)|^2 |\vec{X}_{10}|^2 = \frac{Z_0}{2k^2} \left| \frac{ik^3}{3} \sqrt{2} M'_{10} \right|^2 |\vec{X}_{10}|^2 \\ &= \frac{Z_0}{2k^2} \cdot \frac{2k^6}{9} \cdot \frac{3}{4\pi} \cdot \frac{36}{25} M^2 \beta^2 \cdot \frac{3}{8\pi} \sin^2 \theta = \frac{9Z_0 k^4}{200\pi^2} M^2 \beta^2 \sin^2 \theta \end{aligned}$$

and the total power radiated is

$$P(l, \nu) = \frac{Z_0}{2k^2} |a_{nl}(l, \nu)|^2 = \frac{Z_0}{2k^2} \cdot \frac{2k^6}{9} \cdot \frac{3}{4\pi} \cdot \frac{36}{25} M^2 \beta^2 = \frac{3Z_0 k^4}{25\pi^2} M^2 \beta^2$$